

Creating a Digital Wallet Application in App Inventor 2

This month, we present another interesting app in our App Inventor 2 series. Digital wallets are a trending topic ever since our country decided to opt for a 'less cash' economy. Creating this app will give readers a good insight into how a digital wallet really works.

After reading and practically trying out the apps in this series, many readers have mastered the components available in the palette and have had much fun moving them between the designer and block editor.

Digital wallets or digital payment services are the talk of the town nowadays because of the government's initiatives to go digital and the Digital India campaign. Vendors are opting for mobile payments rather than cash, and this is a smooth as well as hassle-free process. Let's explore the functionality of digital wallet payment systems with the help of a mobile application which we will develop over the course of this two-part tutorial.

Before that, I would like to define what a digital wallet is and how it works. A digital wallet is an application or feature within an electronic device which enables it to make electronic transactions. It has a linked bank account number from which all the transactions are processed and each transaction is recorded for future reference. The device needs to be connected to the Internet to make payments.

The theme of the application

The theme is pretty simple and you might have already come across various mobile payment applications. We will make a similar app, and demonstrate its

debit and credit features in the digital wallet account.

GUI requirements

For every app, we need a graphical user interface or GUI, which helps the user to

interact with the on-screen components. How each component responds to user actions is defined in the block editor section. In this app, we will have multiple screens and will define the look and feel for each of them.

There are four GUI screens. Their names are:

1. Screen1
2. Login Screen
3. New User Screen
4. Purchase Screen

The GUI requirements for the application are listed below.

1. **Label:** Labels are static text components, which are used to display some headings or markings on to the screen.
2. **Button:** Buttons will let you trigger the event and are very essential components.
3. **Horizontal arrangement:** These are special components which keep all child components aligned within themselves. The horizontal component keeps all the child components horizontally aligned.
4. **Notifier:** These are used to display some instructions or to give controls over existing components. You will discover more about their functionality as we implement them in our application.
5. **TinyDB:** This is the internal storage within the device, which will be used to keep confidential data such as passwords, etc.
6. **Text Boxes:** These are pretty simple

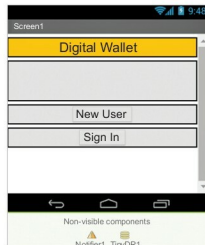


Figure 1: Designer screen (Screen1)

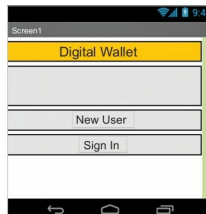


Figure 2: How the application looks

Table 1

	Component name	Purpose	Location
1	Label	To display a label	Palette-->User Interface-->Label
2	Button	To trigger events	Palette-->User Interface-->Button
3	Horizontal arrangement	To arrange the child components	Palette-->Layout-->Horizontal Arrangement
4	Notifier	To display on-screen information	Palette-->User Interface-->Notifier
5	TinyDB	To store data	Palette-->Storage--> TinyDB